Detecting pathological scalp recordings using features of the EEG background pattern

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Abstract: The aim of this study was to optimize the performance of EEG-based pathology decoding methods by incorporating features that capture specific properties of the EEG background pattern. An extensive investigation was conducted, encompassing a diversity of experimental setups, comprising signal inputs, feature representations, feature extraction methods, feature selection criteria, and binary classifiers. Among the various experiments, the optimal results were achieved by utilizing a VGG to derive feature maps from spatial features, yielding specificity and sensitivity values ranging from 77.9 to 85.0% and 75.4 to 76.9%, respectively. The strategy of extracting features from crops of overlapping signals demonstrated comparable performance, with specificity and sensitivity values ranging from 79.1 to 84.0% and 75.2 to 76.8%. The evaluation of various signal inputs, extraction techniques, and selection criteria revealed that the inclusion of information reflecting traits of the posterior dominant rhythm (PDR) was highly discriminative. The results indicated that augmenting a baseline set of cropped features with the proposed features yielded a a marginal, albeit significant, improvement in performance. Our evaluation revealed that random forest outperformed other classifiers, and features derived from channels located in the left temporal/frontal and occipital lobe were more effective in differentiating between pathological and non-pathological states.

1 Introduction

Electroencephalography (EEG) is a widely used and non-invasive method for analyzing and measuring electrical activity in the brain. Its applications have been extensively explored for several decades, providing valuable insights into the complex relationships between cognitive processes, behavioral states, motor function, and neurological disorders (Brazier, 1961). Clinicians rely on EEG to assess brain activity and support the diagnosis of various neurological conditions. The analysis of EEG recordings in clinical neurology usually consists of detecting transients and analyzing the background pattern (van Putten, 2009; Schomer et al., 2012). The distinction between transients, representing rare physiological and pathological events, and the background pattern, describing the mean statistical EEG characteristics, is used to diagnose brain abnormalities. However, the reliable recognition of pathological EEG waveforms requires substantial expertise, and the subjective nature of EEG analysis limits diagnostic accuracy.

To address these limitations, computerized EEG analysis methods have been developed with the aim of automating the detection of EEG abnormalities (Shirrmeister et al., 2017; van Leeuwen et al., 2019; Gemein et al., 2020; Singh et al., 2021). Commonly referred to as pathology decoding methods, these supervised learning solutions offer numerous advantages to clinicians, although current solutions remain insufficient and further research is needed to improve their accuracy. Prior work has predominantly focused on developing feature-based and end-to-end deep Convolutional Neural Network (ConvNet) models trained on one publicly available dataset (Obeid and Picone, 2016). Although deep ConvNets have shown to achieve superior performance, feature-based methods based on binary classifiers have also produced comparable results (Gemein et al., 2020; Singh et al., 2021; Bajpai et al., 2021).

Despite the plethora of ConvNets proposed for EEG pathology decoding, little emphasis has been placed on developing methods utilizing features that capture specific properties of the EEG background activity, despite evidence suggesting changes in this pattern as strong indicators of general brain dysfunction (Juhàsz et al., 1997; Jordan, 2004; Jin et al., 2006; van Putten, 2007; Knyazeva et al., 2008; Spronk et al., 2011). Lodder and van Putten (2012) introduced five key features to quantify the EEG background pattern. van Putten (2007) earlier established four time-frequency images to visualize the time-localized frequency information for a triplet of features quantifying the EEG spatial distribution and coherence. The visualization of these features has already proven to be useful in analyzing and reviewing routine recordings, however, there have been no efforts to integrate them into automated decoding methods.

This study aims to verify the hypothesis that incorporating features quantifying changes in EEG background activity can improve the performance of existing EEG pathology decoding methods. The focus is on the application of the features presented by van Putten (2007) and Lodder and van Putten (2012) in feature-based decoding methods.

The rest of this paper is organized as follows. Section 1.1 summarizes related work in the area of our experimental design. Section 2 encompasses the data acquisition and preprocessing (subsection 2.1), feature design (subsection 2.2), and experimental setup (subsection 2.3). The results of the experiments are detailed in Section 3, followed by the discussion on the interpretation of the results and experimental design in Section 4. The conclusion and recommendations for future work are presented in Section 5.

1.1 Related work

Pathology decoding methods consists of two components: feature extraction and classification. End-to-end methods leverage raw or minimally processed signal data as input, while feature-based methods rely on features to represent the raw signal data. In the literature, various EEG representations have been utilized as inputs. Generic ConvNets require two-dimensional inputs, so papers proposed raw

and feature inputs from multi-channel data, and raw and feature inputs from single-channel data (i.e., a spectrogram taken from a specific channel, or any feature taken from a channel that outputs a two-dimensional shape). The following provides a summary of previous studies, including the model results presented in Table 1.

Lopez et al. (2017) conducted a study that compared the performance of multi-channel and single-channel input data. The EEG signals were analyzed using a TCP montage and Mel-frequency cepstral coefficients (MFCCs) were extracted as features. The authors evaluated the use of different segments of signal data, referred to as crops or epochs, as inputs to the classifiers. Three binary classifiers were employed for decoding the cropped features, including k-nearest neighbors (KNN), random forest (RF), and a deep classifier (MLP). Additionally, principal component analysis (PCA) was proposed to reduce the dimensionality of the cropped features, and a hidden Markov model with Gaussian mixture emissions (GMM-HMM) was introduced to model the cropped features in a probabilistic framework.

Shirrmeister et al. (2017) proposed shallow and deep ConvNets trained on frequency features from multi-channel EEG data in six frequency bands (delta, theta, alpha, low beta, high beta, and low gamma). The ConvNets used the Softmax activation function in the output layer for binary classification. The training data consisted of reduced versions of the EEG signals, ranging from 1 to 16 minutes following the beginning of the recording.

Roy et al. (2018) and Yildirim et al. (2018) validated 1D ConvNets trained on raw single-channel EEG data. Roy et al. (2018) experimented with different channel pairs while Yildirim et al. (2018) specifically used the T5-O1 channel pair from a TCP montage. Additionally, Roy et al. (2018) applied a generic ConvNet trained on images created by Gramian angular fields (GAF), which outputs a temporal correlation computed using polar coordinates, with each element a trigonometric sum between different time intervals.

van Leeuwen et al. (2019) adapted the ConvNet proposed by Shirrmeister et al. (2017) but trained on raw multi-channel EEG data. The methods are essentially similar but differ in minor degree. They were able to assign sleep stage probabilities to consecutive segments of EEG and explored with including contextual information such as age and gender. Before the last classification layer, they added an extra average pooling layer to take an average along the time point axis, enabling them to add new features to the network.

Gemein et al. (2020) followed a cropped-wise feature extraction strategy for multi-channel EEG data, including time, frequency, and connectivity features. Frequency features were computed from crops within specific frequency bands, as earlier outlined by Shirrmeister et al. (2017). The authors used various binary classifiers, such as support vector machine (SVC), auto-sklearn classifiers (ASC), RF, and four deep classifiers (shallow and deep ConvNets, TCN, and EEGNet). They also proposed Riemannian geometry to decode pathology based on the computed covariance matrix for each crop.

Amin et al. (2019) used ConvNets with pre-trained weights (VGG16 and AlexNet) fine-tuned on raw multi-channel EEG data. As opposed to generic ConvNets, these ConvNets have fixed weights specifically designed to extract high-level feature maps from general images. With the flexibility of the network, a desirable output was chosen by re-scaling the output dimensions of the network to extract features. They proposed SVC to classify the feature maps produced by the network.

Singh et al. (2021) used spectograms derived from single-channel data computed using Fourier (STFT), chromograms, and MFCCs. A VGG19 was used to extract feature maps and they proposed to decode them using three classifiers, including logistic regression (LR), RF, and SVC. The F7-STFT combination yielded the best performance among all individual single-channel tests.

Bajpai et al. (2021) followed a similar approach but opted for spectograms based on Fourier and focused on one specific channel. They used the first minute from each recording of the re-referenced T5-O1 channel pair derived from a TCP montage. Previously developed ConvNets for specific and nonspecific EEG classification tasks were proposed to classify the spectogram images, including DenseNet, InceptionNet, and Seizurenet.

Table 1 Overview of papers that proposed computerized solutions for pathology decoding. Spec, Sens, and Acc are in %, *= Softmax classifier, **= validated on TUH Abnormal Corpus (Obeid and Picone, 2016).

Paper	Input	Model	Spec	Sens	Acc
van Leeuwen et al. (2019)	multi *(raw)*	ConvNet-*	90	74.8	-
		ConvNet+Age-*	90	74.3	-
		ConvNet+Sleep-*	90	76.3	-
		ConvNet+LSTM-*	90	71.1	-
Amin et al.** (2019)	multi *(raw)*	AlexNet-SVC	94.7	78.6	-
		VGG16-SVC	94	77.8	-
Yıldırım et al.** (2018)	single *(raw)*	1D-ConvNet-*	-	-	79.3
Roy et al.** (2018)	single *(raw)*	1D-ConvNet-*	-	-	76.9
		1D-ConvNet-RNN-*	-	-	82.2
		MLP-*	-	-	54.1
		LR	-	-	49.1
Bajpai et al.** (2021)	single *(features)*	SeizureNet-SVC	100	90.5	96.7
		InceptionNet-SVC	100	74.6	88.4
		DenseNet-SVC	100	87.3	94.2
Singh et al.** (2021)	single *(features)*	VGG19-RF	86.7	79.5	82.2
		VGG19-SVC	82.9	77.2	79.3
		VGG19-LR	84.4	78.4	80.8
Roy et al.** (2018)	single *(features)*	ConvNet-*	-	-	70.4
		TCNN-RNN-*	-	-	71.5
Lopez et al.** (2017)	single *(features)*	ConvNet-MLP-*	-	-	58
Gemein et al.** (2020)	multi *(features)*	Deep ConvNet-*	91.9	75.9	84.6
		Shallow ConvNet-*	87.9	79.7	84.1
		TCN-*	91.6	79.7	86.2
		EEGNet-*	92.9	72.1	83.4
		RF	88.3	79	84.1
		SVC	92.7	66.7	80.8
		RG	92.7	77.8	85.9
		ASC	88.1	80.6	84.7
Shirrmeister et al.** (2017)	multi *(features)*	Shallow ConvNet-*	81.9	75.4	78.8
		Deep ConvNet-*	94.1	75.1	85.4
Lopez et al.** (2017)	multi *(features)*	RF	-	-	83
		KNN	-	-	68.8
		GMM-HMM	-	-	74.4
		PCA-HMM	-	-	78.8

2 Materials and Methods

2.1 Data acquisition and preprocessing

The study was performed with labeled routine scalp recordings collected by the department of Clinical Neurophysiology at Medisch Spectrum Twente. The EEGs were performed on patients diagnosed with epilepsy. Semi-random patient IDs were assigned to individual recordings to anonymize the data. Standard EEG caps were used with 19 Ag–AgCl electrodes placed according to the international 10–20 electrode system (see Fig. 1). Electrode impedances were kept below 5 kΩ to reduce polarization effects. For most recordings, a standard 20-minute protocol was followed. The EEG signals were measured at a sampling rate of 250 Hz. Recordings were annotated for eyes open, eyes closed, hyperventilation, and photic stimulation.

Recordings not compatible with the 10-20 system were not used and recordings with a sampling rate different than 250 Hz were resampled. Recordings containing less than 5 minutes of signal data were rejected and more than 30 minutes were capped. The first minute of each recording was excluded to accommodate for noise caused by positioning the cap's location on the scalp. The amplitude of the signals was capped at $\pm 800\ \mu V$ to reject unphysiological high values. The distribution of values was examined per 10 seconds and

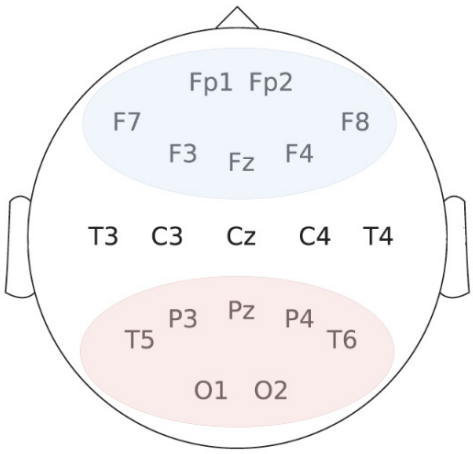

Fig. 1: Modified topographical map of the 19 electrodes placed following the international 10–20 system (Gemein et al., 2020). The blue and red overlays depict the anterior and posterior regions respectively. The electrodes are labeled and distinguishable by letters, located on the frontal, temporal, parietal, and occipital lobes.

segments with a deviation greater than five times the standard deviation were discarded. After preliminary preprocessing, a total of 261 pathological and 692 normal recordings were used. The recordings contained male and female patients with ages ranging from 6 to 103 years.

2.2 *Feature design*

We proposed nine background features, consisting of five quantitative features (detailed in sections 2.2.2-6) and four spatial features (outlined in sections 2.2.7-9). Three of the features capture specific properties of the posterior dominant rhythm (described in sections 2.2.2-3 and 2.2.7), five quantify changes in the power ratio between specific channel pairs (outlined in sections 2.2.4-6 and 2.2.9), and one synchrony measure between neighboring channels (discussed in section 2.2.8). The following sections provide a brief introduction to the background features. A more in-depth formulation of these features can be found in Appendix 8.2.

2.2.1 Spectral estimation: Welch's averaged periodogram method (welch scipy v1.9.1) was used to estimate spectral power. To be more specific, let $V(t)$ be a matrix of a single recording with $c = \{1...N\}$ channels over time t. The signal data retrieved from channel c was divided into half-overlapping segments (256 data points per segment) with a window length (NFFT) of 512, each of which was detrended and windowed using a Hanning window. The frequency resolution of the estimated spectrum is 1/NFFT Hz. Let the output of Welch's averaged periodogram method be denoted as:

$$P(c, j, f) : c \in [1...N], j \in [1...M], f \in [f_{min}, f_{max}]$$

Each entry in $P(c, j, f)$ contains a discrete Fourier coefficient for channel c, segment j, and frequency f. Depending on the features described in the following subsections, either a common reference montage or a small Laplacian was used to analyze the signal data.

2.2.2 Alpha rhythm frequency: Healthy adults commonly experience low amplitude and mixed frequency background rhythms (Lodder and van Putten, 2012). The patient's background rhythm is normally characterized by a dominant alpha rhythm in the posterior region when the eyes are closed and when the patient is in a relaxed state of wakefulness (Niedermeyer, 1997). The frequency of the alpha rhythm increases with age, beginning at about 4 Hz at four months, 6 Hz at six months, 8 Hz at three years, and stabilizing at about 10 Hz at ten years (Eeg-Olofsson et al., 1971; Lindsley, 1939; Niedermeyer, 2011). From this point, the peak frequency may decline slowly depending on patient-specific characteristics (Aurlien et al., 2004). According to an early study by Brazier and Finesinger (1944), the mean peak frequency of 500 adult control subjects was 10.5 $\pm$0.9 Hz. However, Khan et al. (2018) reported intra-individual peak frequency variations of $\pm$0.67 Hz and $\pm$0.46 Hz for 120 patients with focal and generalized epilepsy, including 40 control subjects all aged from 30 to 40. A reference to discriminate between normal and pathological peak frequencies is based on an estimated mean trend of normal control subjects, as previously outlined by Aurlien et al. (2004), van der Stelt (2008), and Segalowitz et al. (2010).

Table 2 Mean reference trend accepted for normal alpha rhythm peak frequencies at different ages.

Age (years)	Frequency (Hz)
0-1	5.3 ± 1.8
2-3	6.8 ± 1.8
4-5	7.9 ± 1.8
6-7	8.7 ± 1.8
8-15	9.5 ± 1.8
16-50	9.9 ± 1.8
>51	9.1 ± 1.8

The alpha rhythm frequency is defined as a dominant peak found in the power spectra estimated from channels in the posterior region. More specifically, the alpha rhythm generator is thought to be located in the occipital lobes. However, it is not uncommon for the dominant rhythm to fluctuate and change in location, amplitude, and width, or not to emerge at all (i.e., absent peaks or multiple insignificant peaks). These fluctuations make it difficult to robustly estimate the underlying alpha rhythm frequency.

We used an iterative curve-fitting method to localized segments of EEG to approximate the two most dominant peak locations, including their amplitudes and widths. We quantified the importance of these peaks with respect to other peaks found using a correlation computation. Based on these findings, clusters were formed and the parameters based on the two largest clusters were used to estimate the alpha rhythm frequency:

$$Q_f(j) = \sum_{i}^{M} f_i w_i \tag{1}$$

with f_i and w_i presenting the estimated frequency and normalized correlation components of the M_j peaks found in a recording. A detailed description of all the steps taken is found in Appendix 8.2.1.

Table 2. provides the reference trend approximated from normal control subjects. If the estimated peak frequency differs $\pm$1.8 Hz from the reference trend, the alpha rhythm frequency is considered to deviate from the norm. The results of the estimations are depicted in Fig. 2.

2.2.3 Alpha rhythm reactivity: Reactivity is characterized by an attenuation of rhythmic activity that occurs after the brain receives an external stimulus after an idle state (Schomer and Lopes da Silva, 2010). Eye openings, auditory inputs, or pain can all be stimulus types. The levels of attenuation vary with age, and lower reactivity may reflect reduced neuronal connectivity and weaker levels of neurotransmissions (Gaál et al., 2010). According to van der Hiele et al. (2007) and Babiloni et al. (2010), dementia is linked to reduced reactivity. The prognostic value of reactivity has been investigated in comatose patients by Logi et al. (2011), Douglass et al. (2002), and Ramachandrannair et al. (2005).

Reactivity is quantified as the difference in alpha power between an idle (relaxed, eyes closed) and non-idle (eyes open) state. Given the estimated peak frequency Q_f, the suppression in alpha power

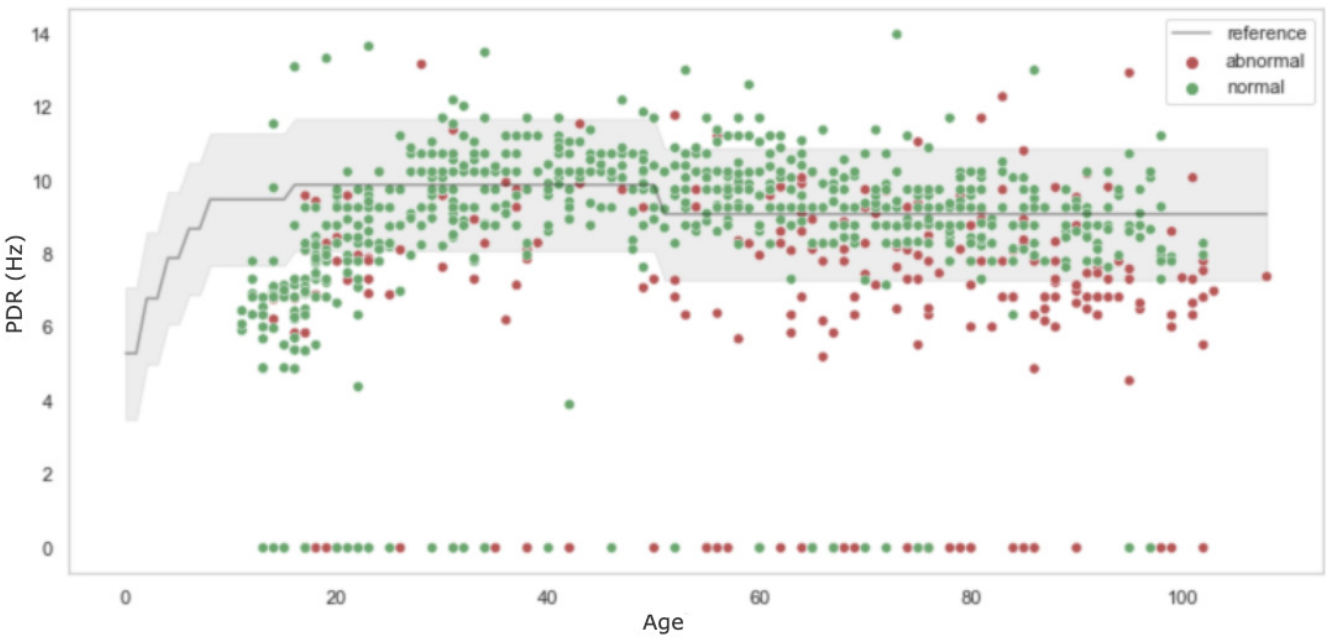

Fig. 2: Alpha rhythm frequency estimation for 953 labeled scalp recordings. The red and green dots denote pathological and non-pathological recordings. 848 recordings showed one dominant peak, 55 had two dominant peaks, and 50 showed no dominant peak (PDR = 0 Hz). Note that a significant number of pathological recordings showed peak frequencies defined well within the mean reference trend.

was measured using a narrow frequency band around this peak. If no peak was found, a frequency band of $f \in [2, 12]$ Hz was used. Let P_{EC} and P_{EO} be the mean occipital power measured in a narrow frequency band on the estimated peak frequency for segments annotated with the eyes closed and eyes open state. The reactivity in alpha power was obtained following:

$$Q_r = 1 - \frac{P_{EO}}{P_{EC}} \tag{2}$$

Reactivity is substantial for $Q_r \geq 0.5$, moderate for $0.1 < Q_r < 0.5$ and low or absent for $Q_r \leq 0.1$. A more detailed description of the definition is found in Appendix 8.2.2.

2.2.4 Alpha power anterio–posterior gradient: Rhythmic activity in a normal awake brain is distributed with an anterior-to-posterior gradient along the scalp (Segalowitz et al., 2010). Schizophrenia and dementia have been linked to abnormalities in the gradient, while drowsiness and deep sleep may cause physiological patterns like slow-wave activity in the posterior region, together with an enhanced alpha and theta activity anteriorly (Stevens and Kircher, 1998; Knyazeva et al., 2008).

A normalized anterio-posterior power gradient was quantified by measuring power in the alpha band for specific channels located in the anterior P_{ant} and posterior P_{pos} regions. Given these mean power estimates, the gradient was defined as:

$$Q_{ap} = \frac{P_{ant}}{P_{ant} + P_{pos}} \tag{3}$$

The gradient is considered within normal range for $Q_{ap} \leq 0.4$, moderately differentiated for $0.4 < Q_{ap} < 0.6$, and abnormal or deviant for $Q_{ap} \geq 0.6$. A more detailed description of the definition is found in Appendix 8.2.3.

2.2.5 Interhemispheric asymmetries: The amplitude of the alpha rhythm is nearly symmetrical for the left and right hemispheres although patient variability exists (Segalowitz, 2010). Brains lesions have been linked to asymmetries between hemispheres (Agius Anastasi et al., 2017; Jordan, 2004; van Putten, 2007). According to Maulsby et al. (1968), approximately 60% of healthy adults have asymmetrical amplitude differences, from which 17% have amplitude differences greater than 20%, and 1.5% have amplitude differences greater than 50%. Based on these findings, interhemispheric amplitude differences greater than 50% are considered pathological. However, it is possible to observe asymmetries of >50%. Such findings are common in patients with asymptomatic EEG characteristics (Ebersole and Pedley, 2003).

Asymmetries were quantified by comparing the alpha power ratio between channel pairs in the corresponding hemispheres. Let LR_{avg} be the average power ratio for a given asymmetry pair $C_{\{LR\}}$. A single asymmetry value was obtained following:

$$Q_s(C_{\{LR\}}) = \underset{f=\{2...12\}Hz}{mean} \left[LR_{avg}(C_{\{LR\}}, f) \right] \tag{4}$$

As a rule of thumb, pathological asymmetry values were defined by a difference in frequency exceeding 0.5 Hz. A more detailed description of the definition is found in Appendix 8.2.4.

2.2.6 Diffuse slow-wave activity: The presence of diffuse slow-wave activity is uncommon in awake and healthy individuals (Ebersole and Pedley, 2003). Normal variants exist, but the presence of slow-wave activity is usually indicative for pathology. Diffused slowing is defined as increased power in theta and delta bands, while power in the alpha and beta bands is attenuated. Sedatives and anesthetics may cause physiological patterns of diffuse slow-wave activity (Cloostermans et al., 2011; San-juan et al., 2010; Blume, 2006).

Diffuse slow-wave activity is present if the mean power has less than one-third of power located above 8 Hz. Using this as a guideline, a mean spectrum was calculated, and the normalized power ratio between P_{low} and P_{wide} was obtained. The discrete power coefficients in the low and wide spectral bands were summed to find

a ratio that quantifies the degree of slow-wave activity:

$$Q_d = \frac{P_{low}}{P_{wide}} \tag{5}$$

Pathological slow-wave activity is present for $Q_d \geq 0.6$ (i.e., less than 40% of power is above 8 Hz). For $Q_d < 0.6$ the recording is considered to have sufficiently fast rhythmic activity. A more detailed description of the definition is found in Appendix 8.2.5.

2.2.7 *Spatial distribution of power in the occipital lobe:*

Using a common reference montage, let $PWR(j)$ be the occipital power measured at channel $O1$ for segment j. A power spectrum was estimated using Welch's method for each localized segment annotated with the eyes closed and eyes open state. Segments annotated during photic stimulation and hyperventilation were not used. Let this be denoted as:

$$PWR(j) = \underset{f=\{2...25\}Hz}{mean} \left[\underset{c \in O1}{mean} \left[P(c, j, f) \right] \right] \tag{6}$$

Each entry contains a list of discrete Fourier coefficients for channel c and segment j. The individual coefficients per segment were concatenated to output the spatial distribution of power in the occipital lobe.

2.2.8 *Spatial distribution of nearest neighbor coherence:*

The synchrony feature was quantified through the calculation of coherence between each electrode position and its nearest neighbors using a coherence function (coherence scipy v1.9.1). For instance, the coherence at channel Cz was computed as the average of four coherence values: coh(Cz,Fz), coh(Cz,C4), coh(Cz,C3), and coh(Cz,Pz). The 19 resulting mean coherence values were further reduced by only using their maximum values (mNNC). Coherence was computed using half-overlapping windows of length 512 after re-referencing to a small Laplacian. The resulting feature is a time- and frequency-dependent function with coherence values ranging from 0 to 1 for different time intervals.

2.2.9 *Spatial distribution of the power gradient:*

The spatial distribution of power along the scalp was quantified by computing the gradient of spectral power across 19 EEG channels, separated into two axes (left-right and up-down). Welch's method was applied to the signals after re-referencing to a small Laplacian, resulting in the estimation of discrete Fourier coefficients A_{ij} with $i = \{1...M\}$ channels and $j = \{1...N\}$ coefficients. The coefficients were weighted based on their Euclidean distance from the reference channel (Cz) in both the x- and y-directions, dx_i and dy_i, respectively. This allowed for the calculation of the center-of-gravity (COG) of spectral power in the x- and y-directions as a function of frequency j.

$$COG_X(j) = \sum_{i=1}^{M} \frac{A_{ij} dx_i}{A_{ij}} \tag{7}$$

$$COG_Y(j) = \sum_{i=1}^{M} \frac{A_{ij} dy_i}{A_{ij}} \tag{8}$$

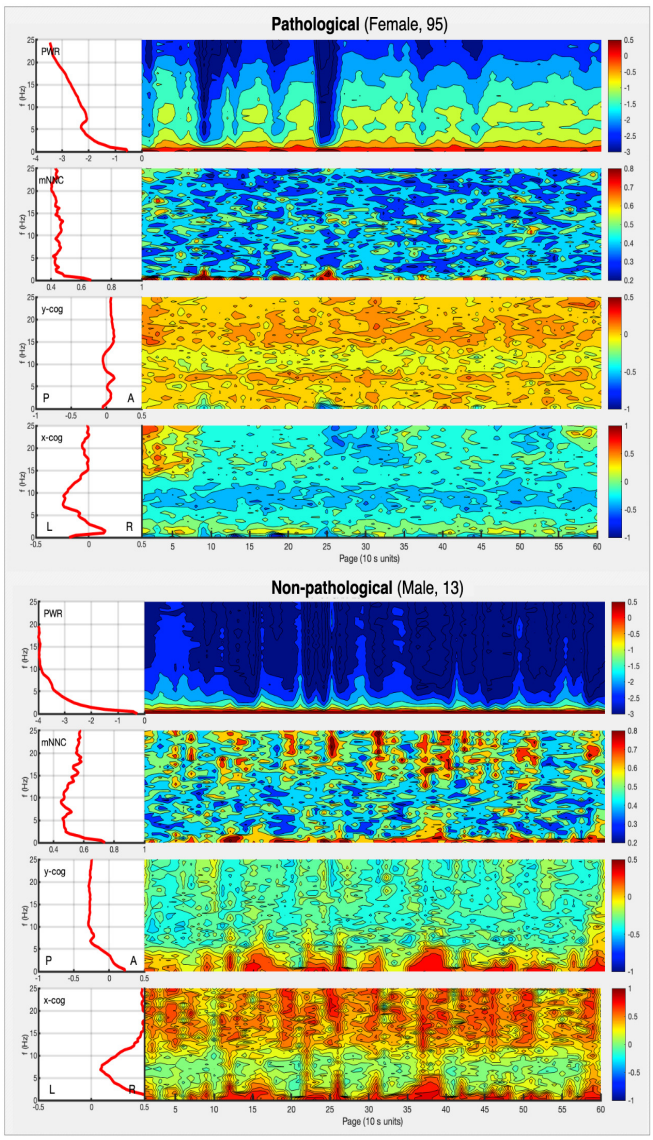

Fig. 3: Comparison of two EEG recordings displayed in a time-frequency visualization of three spatial features. The upper plot highlights a pathological recording, showing a wide peak frequency at 8 Hz (PWR) and weak signal coherence (mNNC) along with an evenly distributed power gradient over the scalp (y-cog and x-cog). The lower plot showcases a non-pathological recording with a peak frequency of 7 Hz, exhibiting weak signal coherence and more power concentration in the posterior and right regions.

2.3 *Experimental setup*

Fair comparisons to strong baselines are essential to compare experiments and to assess the robustness and reliability of the decoding results. However, the diversity in channels, input lengths, and features employed for decoding pathology poses a challenge in identifying the precise factors that substantially contribute to the efficacy of a model versus those that do not. Lopez et al. (2017), Shirrmeister et al. (2017), and Gemein et al. (2020) proposed various computational features extracted from overlapping crops of multi-channel inputs, while Roy et al. (2018), Singh et al. (2021), and Bajpai et al. (2021) extracted features from spectograms derived from single-channel inputs. Although these papers have in common that they use features to represent the raw signal data, their strategies to extract relevant features are inherently different.

We used two feature extraction approaches based on multi- and single-channel inputs. In both experiments, we trained models on reduced versions of signal data by using only the first 2, 4, 8, and

Table 3 All features computed within the multi-channel feature extraction procedure sorted by feature domain. Feature domains include frequency, time, background, patient information, and connectivity.

(CWT/DWT)	**(DFT)**
Bounded variation	Maximum
Maximum	Minimum
Minimum	Mean
Mean	Power
Power	Power ratio
Power ratio	Spectral entropy
Spectral entropy	Value range
Variance	Variance
(Time)	
Maximum	Hjorth activity
Minimum	Hjorth mobility
Mean	Hjorth complexity
Median	
Skewness	Zero crossing
Kurtosis	Zero crossing derivative
Line length	Higuchi fractal dimension
Flat spots	Petrosian fractal dimension
Lumpiness	
Energy	SVD entropy
Nonlinear energy	SVD Fisher information
(Background)	**(Patient information)**
Alpha rhythm frequency	Age
Alpha rhythm reactivity	Gender
Alpha power anterio–posterior gradient	
Interhemispheric asymmetries	**(Connectivity)**
Diffuse slow-wave activity	Phase locking value

16 minutes following the beginning of a recording, as outlined by Shirrmeister et al. (2017). Furthermore, we adopted a feature selection procedure to select discriminative features before classification. Our experiments utilized three binary classifiers, including support vector classifier (SVC), logistic regression (LR), and random forest classifier (RF) as previously proposed in the literature.

2.3.1 Multi-channel feature extraction: We adopted a cropped-wise strategy to extract features from overlapping crops of input data, as outlined by Gemein et al. (2020). We computed power estimates for unique channels within the following frequency bands: 0–2 Hz, 2–4 Hz, 4–8 Hz, 8–13 Hz, 13–18 Hz, 18–24 Hz, 24–30 Hz, and 30–50 Hz. Gemein et al. (2020) reported superior performance using a Blackman-Harris window with 50% band overlap. In line with their suggestion, we used the same window function with 50% band overlap to weigh all segments. For each power estimate produced several statistical properties were computed. Time-complexity features were computed for each crop. Phase locking values were computed for unique channel combinations within each crop. The dimensionality of the cropped features was reduced following a median aggregation over segments. Also the patient's age was included and gender was added as binary variable. After computing the cropped features, we performed a feature selection procedure to select the most discriminative feature values. This procedure is further detailed in subsection 2.3.2. After the initial feature selection, we added twelve feature values from five background features, including four specific values and eight asymmetry values.

A summary of the cropped features is presented in Table 3, and the decoding strategy and time-complexity feature definitions can be found in Appendices 8.1.1 and 8.3, respectively.

2.3.2 Multi-channel feature selection: A feature selection procedure (selectkbest scikit-learn v1.1.2) was applied to identify discriminative features prior to classification. The procedure utilized univariate statistical tests to quantify the relative variance of K-selected features and their associated labels. An iterative approach was adopted to determine the optimal number of features, K, by successively removing one feature from the initial set, evaluating the performance of a classifier, and selecting the set of features that resulted in the highest accuracy. We used LR due to its limited number of hyperparameters.

We employed both F-test and mutual information statistics for feature selection, as these methods are commonly utilized for numerical input data when the target variable is categorical. The F-test is based on a linear dependency estimation between two random variables, while mutual information is non-parametric and requires a large sample size to accurately estimate statistical dependency. Although no significant difference was observed, we opted for the F-test as it produced slightly better results.

The initial computation generated 8943 feature values, and a baseline set of 5236 values was selected through feature selection.

2.3.3 Single-channel feature extraction: We employed the strategy of extracting feature maps from spatial features using a VGG. The F7-STFT combination was found to provide the best decoding performance according to Singh et al. (2021), but we opted to use the raw input from the O1 channel derived using a common reference montage based on previous findings from Yildirim et al. (2018) and Bajpai et al. (2021). We computed three spatial features, described in subsections 2.2.7-9, which are visualized in Fig. 3.

Feature maps were extracted using a VGG19. The VGG19 was retrieved from Tensorflow (v2.10.0) and the pre-trained weights were used from imagenet. We compiled the network using Adam stochastic optimizer with a categorical cross-entropy function to estimate the loss. Singh et al. (2021) and Bajpai et al. (2021) did not specify at which layer features were extracted, so we assumed they were extracted at the last max pooling layer. For the VGG19 this corresponds to an output shape of (7x7x512). Before conducting our experiments in a variety of configurations, we used an intermediate step and experimented with extracting features at various output layers. The specific steps of this approach are further detailed in subsection 2.3.4.

For each spatial feature, we computed 953 feature maps with dimensions (953x14x14x512). A more detailed definition of the decoding strategy is found in Appendix 8.1.2.

2.3.4 Single-channel feature selection: To find the best output layer for our main experiments, we extracted feature maps from layers (56x56x256), (28x28x512), (14x14x512), and (7x7x512). Principle component analysis (PCA) was used to find the best balance between noise and discriminative information for feature maps extracted at each layer. To implement the approach, a scree plot was utilized to visually identify a range of potential components for PCA. PCA was fitted iteratively by selecting N components within the range and transforming the feature maps accordingly. LR was trained on the feature maps, and performance was measured using accuracy. The number of components with the best performance was selected. Empirical results showed that feature extraction at higher layers combined with dimensionality reduction outperformed extraction at the lowest layer. However, higher layer extraction resulted in more components, with no clear separation in components and the variation they described. Extraction at (14x14x512) achieved the best balance between component separation and decoding performance.

After flattening the feature maps, initial extraction produced 100352 feature values. The number of components for PWR was set to 25, yielding 23775 values; COGx and COGy were set to 29 and 31, yielding 27579 and 29481 values, respectively; and mNNC was set to 38, yielding 36138 values.

As an alternative approach, we combined the feature values of the best-performing spatial features to create a surrogate model and selected only the values recommended by the selection procedure outlined in subsection 2.3.2.

2.3.5 Importance analysis of features: To quantify the contribution of different features to our experiments, we utilized the feature importance property of RF. By calculating the purity of a data split that is obtained through a feature, RF determines the importance of the feature. As a rule of thumb, the purer the data split and

the earlier the feature is included in the tree, the higher its importance. This allowed us to quantify the significance of each feature in the decision-making process. We used permutation feature importance (permutation_importance scikit-learn v1.1.2), as it offers an unbiased evaluation compared to impurity-based feature importance, and it is not influenced by high-cardinality features. Feature importance was measured as the mean decrease in accuracy, with the most important features causing the highest decrease in accuracy. To aid in visualizing the feature importance, we assigned textual labels to individual feature values and created topological plots to highlight the most important features within specific frequency ranges and electrode locations.

2.3.6 Model validation: To ensure a fair comparison between models, 80% of the recordings were used for training and 20% for testing. For the test set an equal number of pathological and non-pathological recordings were selected. To account for potential biases in the chosen sets, the recordings were randomly divided (i.e., subsampling with replacement) into 1000 sets of train and test. Model performance was validated using 8-fold cross-validation, with predictions made on the left-out test set and compared to the ground truth. Predictions were mapped to binary values and categorized into four categories, with decoding performance quantified in specificity and sensitivity ratios:

$$\text{Specificity}(\%) = \frac{TN}{TN + FP} \cdot 100\%$$

$$\text{Sensitivity}(\%) = \frac{TP}{TP + FN} \cdot 100\%$$

with TP, FP, TN, and FN denoted by the number of true positives, false positives, true negatives, and false negatives, respectively.

McNemar's test (mcnemar statsmodels v0.13.2) was used to compare prediction errors between experiments. The hypothesis was formulated as follows:

$H0$: prediction errors are significantly different.
$H1$: prediction errors are not significantly changed.

The hypothesis was tested using a 2x2 contingency table, with the contingency values determined based on the specificity and sensitivity of the model predictions. If both models were correct, one value was recorded as "yes/yes". If one prediction was correct and the other incorrect, one value was recorded as "yes/no" or "no/yes", and if both were incorrect, one value was recorded as "no/no". The significance of the prediction errors of the classifiers was determined by comparing the p-value to H_0. If the p-value was less than 0.05, it meant that the prediction errors differed significantly and the null hypothesis was rejected. A continuity correction was applied to the calculation.

2.3.7 Reproducibility of experiments: The experiments were conducted on a Windows OS computer with an AMD Ryzen 7 3700X 8-Core Processor running at 3.60 GHz and equipped with an NVIDIA RTX 3080 Ti GPU. Python 3 was used to implement the computational features and classifiers, while the spatial features were calculated using Matlab. The classifiers were retrieved from the Scikit-learn (v0.21) and their specific parameters were optimized through a grid-search approach.

The study relied on an adapted version of the open-source toolkit Brainfeatures, which has been optimized and made accessible on (github.com/RobvanderNagel94/pathologylib).

3 Results

The decoding results can be found in Table 4, which is outlined in Appendix 8.4. Fig. 3 visually presents the bootstrapped specificities for each model trained on features extracted from reduced signal data. Table 4 and Fig. 3 indicate a considerable variation in performance across different minutes of extraction. The experiments showed that performance increased from 4 to 8 minutes, and decreased after 8 minutes considering different features, classifiers, and extraction techniques. Furthermore, all decoding models were found to have a higher ratio of false negatives than false positives, meaning that the models were more likely to classify pathological examples as non-pathological rather than non-pathological examples as pathological. These results align with those of Lopez et al. (2017), Schirrmeister et al. (2017), Gemein et al. (2020), and van Leeuwen et al. (2019).

3.1 Muli-channel experiments

The results for multi-channel signal extraction resulted in mean specificities ranging from 76.9 to 84% and sensitivities ranging from 73.0 to 78.0%. The deviation in specificity and sensitivity values observed in our experiments ranged from 0.94 to 1.28. RF achieved the highest performance with mean specificity and sensitivity levels of 84.0% and 78.0%, respectively. An increase in performance was observed when the number of minutes used to extract features was increased, with an optimum achieved at 8 minutes. Performance declined when using 16 minutes.

We evaluated the impact of individual background features on the predictions and compared them to the baseline performance. To this end, we conducted experiments using the baseline set of cropped features as described in subsection 2.3.2 and added the output of each background feature individually. The results from the five one-to-one comparisons showed no significant differences. However, when all background features were combined, significant improvements were observed. Further analysis revealed that among the 1000 runs, the highest p-value obtained was 0.092 and approximately 82.0% of the configurations had p-values below 0.05. The cross-validation results showed a slight improvement in specificity, with an average increase of 0.11% across all configurations.

Moreover, we examined the impact on model performance by transforming the background features into binary values using specific thresholds outlined in subsections 2.2.2-6. However, this approach resulted in a substantial decrease in performance across all configurations, with a mean decrease in specificity of 0.09%. Further analysis revealed that some values were misclassified, specifically in cases where patients were labeled as pathological despite their alpha rhythms being classified as non-pathological.

3.2 Single-channel experiments

When performing feature extraction from spatial features, the best results were obtained by computing the PWR feature with 8 minutes of signal data in conjunction with an RF classifier. This specific configuration resulted in specificity and sensitivity values of 85.0% and 78.0%, respectively. No other configuration outperformed these results, with SVC producing specificity and sensitivity values of 81.0% and 78.0% and LR yielding 83.2% and 77.3%. Compared to the PWR feature, using the COGx and COGy features for extraction showed a mean decrease of 4 to 2.6% for extraction at 2 minutes, 2.9 to 1.6% at 4 minutes, 4.5 to 3.7% at 8 minutes, and 1.1 to 2.1% at 16 minutes. The mNNC feature performed significantly worse compared to all other features.

Of all the classifiers tested, RF outperformed the others, regardless of the features and extraction techniques used, while LR produced similar results but with a small decrease in performance. SVC performed the poorest among all classifiers in different configurations, which aligns with the findings of Gemein et al. (2020), who showed that all feature-based models were congruent for bootstrapped accuracies, except for SVC.

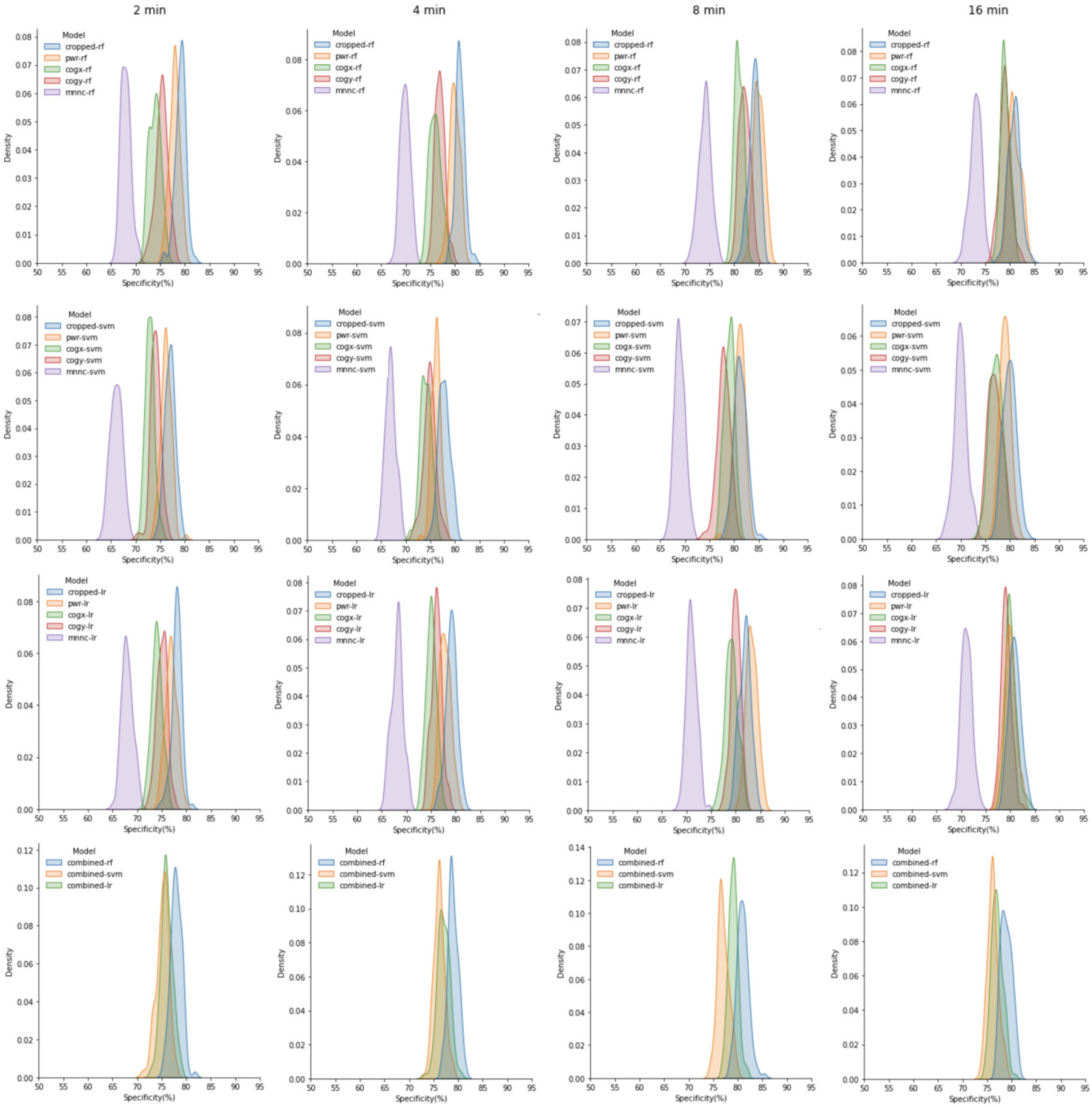

Fig. 4: Performance visualization of bootstrapped specificities for classifiers shown through plots of kernel density estimations of specificity values obtained from validating each model on 1000 sets of train and test data. The features were computed on reduced versions of signal data with varying durations (2, 4, 8, and 16 minutes).

3.3 Feature importance

Our initial approach involved quantifying feature importance through variance-based feature selection before classification. We grouped features by domain (frequency, time, connectivity, patient-specific, and background) and visualized their importance values. Frequency and connectivity features showed higher importance across all extracted frequency bands. Further analysis showed higher variance in delta (0-4 Hz), theta (4-8 Hz), and alpha (8-14 Hz) bands compared to higher frequency bands. Although a significant number of features were computed, only two background features, the anterior-posterior gradient, and the alpha rhythm frequency performed among the top 100.

We validated our initial findings using permutation feature importance. The most important features were the patient's age and gender, followed by various frequency, connectivity, and time features. The most significant background feature was the anterior-posterior gradient, followed by the alpha rhythm frequency, reactivity, diffuse slowing, and various asymmetry values. We averaged the feature importance values for all experiments and plotted them on topographical maps for different frequency bands (as seen in Fig. 4). The delta, theta, and alpha frequency bands were found to be more discriminative for channels T3, T4, C3, and O1, while connectivity showed that channels T4, F7, and F3 were more discriminative for features extracted at higher frequency bands. The channels with the strongest discriminatory power across all frequency bands were T3

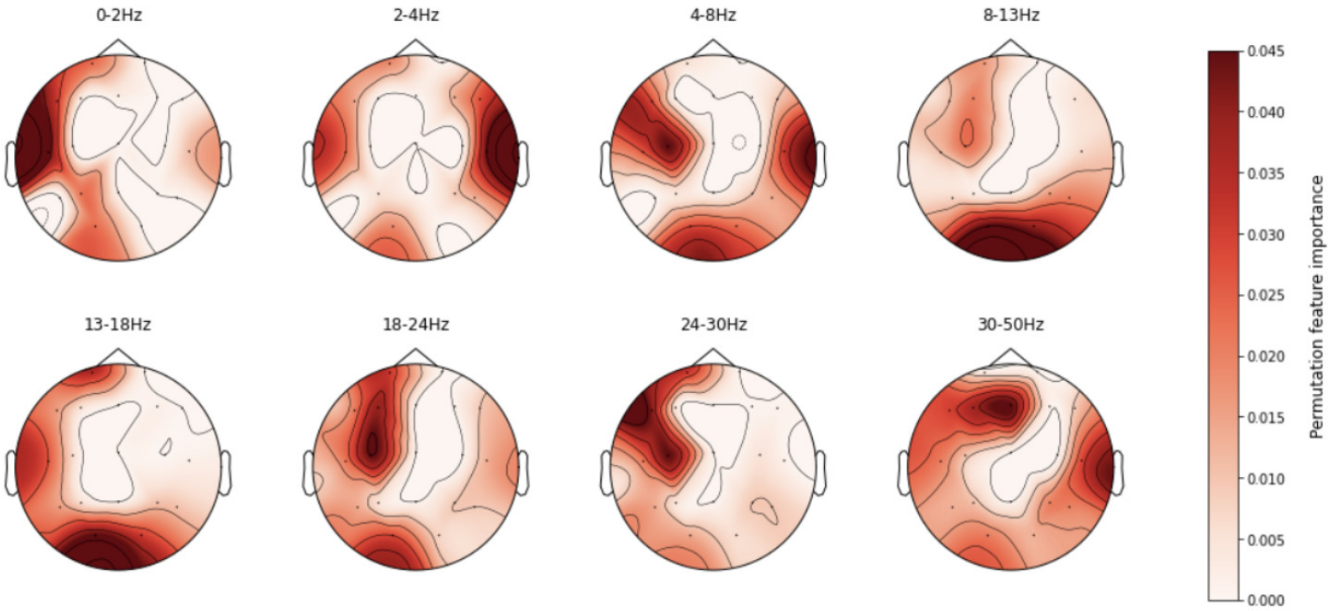

Fig. 5: Feature importance results for all extracted features within different frequency bands. The feature importances were computed by taking the mean of all experiments conducted. The topographical maps indicate that for electrodes T3, C3, and T4, the most informative features were found between 0-8 Hz. Electrodes O1 and O2 showed higher importance for features within the 8-18 Hz range. For electrodes T4, F7, and F3, features extracted in higher frequency bands demonstrated higher importance.

and those in the occipital region.

4 Discussion

4.1 Experimental setup and limitations

Although this paper supports the claim that the performance of feature-based methods can be improved using features of the EEG background pattern, the interpretation of the results should be taken with care. In extent to configurations previously reported in the literature, this study covered:

- **Channel inputs**: multi- and single-channel inputs, reduced versions of signal input lengths.
- **Features**: computational features and spatial features.
- **Feature extraction techniques**: cropped-wise feature extraction and feature extraction using a VGG.
- **Feature selection criteria**: variance-based feature selection and feature selection based on principle component analysis.
- **Classifiers and ensembles**: logistic regression, support vector machine, and random forest.
- **Performance validation**: mcnemars test for ono-to-one comparisons and bootstrapping to quantify the deviation of prediction errors among different configurations.

Despite the variety of configurations examined, the main finding was that models performed within a range of 72-85%, which is lower than previously reported in papers from Gemein et al. (2020) and Singh et al. (2021).

With respect to other papers, our dataset contained significantly fewer recordings than the reference TUH Abnormal Corpus (v2.0.0 contains approximately 1488 pathological and 1529 non-pathological recordings). In addition, there was a disparity in the proportion of labeled recordings (261 pathological and 692 non-pathological). Even so, due to the relatively higher proportion of younger patients in our dataset, the patient heterogeneity may have posed a limitation. Despite not having access to EEG reports, we discovered that the majority of pathological recordings were produced by a small group of patients. Shirrmeister (2017), van Leeuwen (2019), and Gemein et al. (2020) trained models on patients within specific age ranges, as their datasets comprised much more patients, and also the patient heterogeneity was substantial. However, we were not able to do this so the disproportional distribution of patients could reflect in models that performed suboptimal.

In our validation configuration, the distribution of pathological and non-pathological recordings was set equally in the test set to ensure an even comparison among experiments. However, due to this constraint, the sampling technique might not have worked optimal as only a few pathological recordings were available. This resulted in less variation among the various sub-samples in the test sets. In turn, this has an effect on the percentage of recordings used for training as well. Therefore, the test set may not be sufficiently representative to fully test the models' ability to generalize to unseen recordings, even when sampling was considered.

Furthermore, the preprocessing steps used in this study were kept relatively simple, mainly aimed at rejecting unphysiological high values, though subtle artifacts may not have been detected. Even after our initial preprocessing phase, some recordings still showed significant outliers in higher frequency bands. Developing more advanced techniques for artifact removal may have helped to identify and eliminate less obvious artifacts.

4.2 Data quality

Label noise, as stated by Gemein et al. (2020), has a significant effect on decoding performance. To evaluate against possibly noisy labels in our test set, low inter-rater agreement limits the potential decoding performance. Unfortunately, we lacked access to any rater-independent ground truth. Inter-rater agreement in binary EEG classification is reportedly between 86% and 88% (Houfek and Ellingson, 1959). The decoding results, as seen here and in previous studies by Schirrmeister et al. (2017), Leeuwen et al. (2019), and Gemein et al. (2020), seem to align with these numbers and may have reached close to the theoretical maximum due to label noise.

We found that both the consistency of labeling following the annotation requirements and the duration of annotated segments per recording were unevenly distributed. Robust estimations of alpha rhythm frequency, reactivity, and diffuse slowing rely heavily on the availability of annotated segments in the EEG recordings. To minimize the number of rejected recordings, we set a minimum

requirement of two minutes of annotated segments. However, estimations may still be inaccurate for recordings that only meet this minimum requirement. Recordings that were rejected based on this criterion were manually checked and adjusted using the PWR feature's visual output. Our label criteria was based on the definition by Schomer and Lopes da Silva (2010) and involved annotating segments with low frequency activity around the alpha band as "eyes open" and those with high frequency activity as "eyes closed". However, evaluating the robustness of these estimates is challenging as it requires visual inspection of the raw EEG data, making adjustments to the annotations by non-experts unreliable.

4.3 Feature computation and selection criteria

The method used for selecting relevant features in this study involved heavily on the application of dimensionality reduction techniques, such as PCA and variance-based feature selection.

The ANOVA feature selection approach was limited as it only compared one-to-one features and disregarded the contribution of a selection of features. Permutation feature importance results also showed that not all high-scoring features were necessarily deemed discriminatory. The comparison between F-test and mutual statistics produced different results, leading us to choose F-test as it produced better results in this particular setting. However, it is important to note that this may not always be the case for other settings. Additionally, permutation feature importance was only performed on RF and no other validation was done on other classifiers. This means that we were unable to validate if the selected features were truly discriminative across classifiers.

Moreover, in the single-channel experiments, PCA was also utilized to strike a balance between discriminative information and noise. Although the results of the experimentation indicated that the optimal combination was obtained for extraction at the second lowest level, the criteria for determining the number of components may not have been precise, as it relied on the elbow method. Similarly, the elbow method was employed to determine the number of clusters for the alpha rhythm frequency components. In a recent publication by Schubert (2022), it was underscored that the elbow method heuristic, commonly used for this purpose, may result in erroneous conclusions. He stated that it is crucial to exercise caution while utilizing the elbow method and instead opt for superior alternatives such as the variance-ratio criterion of Calinski and Harabasz (1974) or the Bayesian Information Criterion (BIC).

4.4 Stability of classifier-specific parameters

We evaluated the stability of classifier-specific parameters in various configurations to validate the reliability of the decoding results. LR and SVC had the least number of adjustable parameters, whereas RF had greater flexibility in terms of the parameters that could be used to train a decision tree, such as the maximum tree depth, splitting criteria, and number of estimators. This made RF more prone to overfitting due to the greater variation in parameters that could be used. To determine the optimal parameters for RF, we set the maximum tree depth to a range between 2 to 15 and the number of estimators between 1 to 20. Our results showed that the highest tree depth for RF trained on cropped features was 9, with most configurations having values between 4 to 8. The tree depth used when training on the features extracted from the PWR feature ranged from 3 to 9. The number of estimators used for different experiments varied between 2 and 5. Overall, we observed a higher degree of parameter variation when extracting features from 2 to 4 minutes and 8 to 16 minutes.

Our results indicated that the COGx, COGy, and mNNC features required higher tree depths for moderate performance and had greater parameter variations. In particular, the mNNC feature required a significantly higher number of estimators and large tree depth, indicating the potential for overfitting.

5 Conclusion

In this study, we evaluated the use of features suggested by van Putten (2007) and Lodder and van Putten (2012) to enhance the performance of feature-based pathology decoding methods. The best results were obtained by training a RF classifier on features extracted from the PWR feature derived from the O1 channel, resulting in specificities of 77.9 to 85.0% and sensitivities of 75.9 to 76.9% for various signal input lengths. Similar results were found using the strategy of extracting features from overlapping segments of EEG, with specificities and sensitivities ranging from 79.1 to 84.0% and 75.2 to 76.8%, respectively.

The PWR feature showed the highest discriminatory power compared to other spatial features (COGx, COGy and mNNC). In line with the results of Yildirim et al. (2018) and Bajpai et al. (2021), the results also indicate that using features that capture spatial properties of the PDR is effective in distinguishing between pathological and non-pathological recordings. Additionally, the inclusion of five background features in the baseline set resulted in a marginal, albeit significant, improvement in performance, with the anterior-posterior gradient and alpha rhythm frequency being the most impactful features.

Finding the optimal balance between noise and discriminatory information is an ongoing challenge in this field. ConvNets are capable of extracting complex patterns, but this does not guarantee that the resulting model will produce clear, interpretable decoding decisions, particularly in clinical neurology where model interpretability is crucial this might pose an issue.

In conclusion, the field of pathology decoding has significant potential for improvement, particularly in finding discriminatory information and exploring more background features. Advanced feature extraction techniques such as SIFT, SURF, HOG, LBP, and autoencoders could enhance the extraction of discriminatory information for spatial features. It is important to note that while ConvNets are frequently used for image classification, they are not always the ultimate solution due to their complexities in optimization and fine-tuning, as well as challenges in enhancing data. In contrast, feature-based approaches provide greater flexibility and feature interpretability, resulting in models that are easier to understand and explain. These innovative advancements offer promising prospects for automating EEG analysis and revolutionizing clinical practices to better serve the needs of those impacted by neurological disorders.

6 Declaration of competing interest

The authors declare no competing financial interests that could have appeared to influence the work reported in this paper.

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8 Appendices

8.1 Appendix A: Definition of decoding strategies

8.1.1 Multi-channel decoding strategy: Let V_i be a matrix containing all feature values for $i \in I$ recordings. Each row in V_i consist of a feature vector with an equal amount of computed feature values per analyzed segment:

$$V_i = \begin{bmatrix} X_{11} & X_{12} & ... & X_{1M} \\ X_{21} & X_{22} & ... & X_{2M} \\ ... & ... & ... & ... \\ X_{N1} & X_{N2} & ... & X_{NM} \end{bmatrix}$$

with X_{NM} a numerical value for $C_i = \{1...N\}$ segments and $F = \{1...M\}$ feature values. The dimension of the matrix is denoted as $V_i \in \mathbb{R}^{C_i \times F}$. A median aggregation function was used to retrieve one median feature value from all segments. Thus, a column-wise aggregation was performed to retrieve one feature vector per recording. This process was repeated for each recording such that all computational feature values were stored as one aggregated feature matrix $V_{agg} \in \mathbb{R}^{I \times F}$ with corresponding labels $L = \{l_1...l_I\}$ for $l \in [0, 1]$.

8.1.2 Single-channel decoding strategy: Let V_i be an image for recording $i \in I$ produced by a given spatial feature:

$$V_i = \begin{bmatrix} X_{11} & X_{12} & ... & X_{1M} \\ X_{21} & X_{22} & ... & X_{2M} \\ ... & ... & ... & ... \\ X_{N1} & X_{N2} & ... & X_{NM} \end{bmatrix}$$

with $0 \leq X_{NM} \leq 255$ a pixel value for $\{1...N\}$ frequency values and $\{1...M\}$ time values. By design, the VGG19 requires a three dimensional input. The images were first rescaled to $N = 224$ and $M = 224$. Following the RGB coordinate system, we then transformed V_i into a three dimensional shape. We extracted feature maps with output dimension (14x14x512). The feature maps were flattened before being used for classification. Thus, four separate datasets of feature maps were created with corresponding labels $L = \{l_1...l_I\}$ for $l \in [0, 1]$.

8.2 Appendix B: Definition of EEG background features

8.2.1 Alpha rhythm frequency: Using a common reference montage, a power spectrum was computed for channels $c \in \{O1, O2\}$ separately within frequency band $f \in [2, 18]$ using only segments annotated with the eyes closed state $j \in J_{EC}$. For clarity, the following procedures are described for one power spectrum $P(f)$. The estimated power spectrum was log transformed:

$$P_{log}(f) = log(P(f))$$

Dominant peak location: A curve P_{curve} was constructed using three components: two peak components P_{pk1} and P_{pk2} and background component P_{bg}. P_{curve} was fitted to P_{log} and parameters were optimized in a nonlinear least-squares way following the Levenberg–Marquardt method (least_squares scipy v1.9.1).

$$P_{log}(f) \approx P_{curve}(f) = P_{pk1}(f) + P_{pk2}(f) + P_{bg}(f)$$

$$P_{pk1}(f) = A_1 \cdot exp\left(\frac{(f - f_1)^2}{\Delta_1^2}\right)$$

$$P_{pk2}(f) = A_2 \cdot exp\left(\frac{(f - f_2)^2}{\Delta_2^2}\right)$$

$$P_{bg}(f) = B - C \cdot log(f)$$

Parameters A_1 and A_2 are the amplitudes, f_1 and f_2 the center frequencies and Δ_1 and Δ_2 the widths. C is a power-law approximation and B a normalization factor. Parameters were optimized in an iterative way to prevent localized optimizations of components from interfering with the approximation of P_{curve}. Initial values for B and C were set to zero and the parameters were approximated by fitting the background component to P_{log}, while minimizing the error:

$$\{B, C\} = \underset{f \in [2,18]}{argmin} |P_{log} - P_{bg}|$$

After obtaining parameters B and C, parameter C was kept fixed. Initial values for A_1 and f_1 were approximated (find_peaks and peak_prominences scipy v1.9.1) to find the most dominant peak's location and it's amplitude in $(P_{log} - P_{bg})$, while Δ_1 was set to 1. One peak was fitted to P_{log}:

$$\{A_1, f_1, \Delta_1, B\} = \underset{f \in [2,18]}{argmin} |P_{log} - P_{pk1} - P_{bg}|$$

In a similar fashion, initial values for A_2 and f_2 were set based on the second most dominant peak found in $(P_{log} - P_{pk1} - P_{bg})$, while Δ_2 was set to 1. The second peak was fitted to P_{log}:

$$\{A_2, f_2, \Delta_2, B\} = \underset{f\in[2,18]}{argmin}|P_{log} - P_{pk1} - P_{pk2} - P_{bg}|$$

During the iterative process, two evaluations were performed to verify whether P_{curve} was accurately approximated. Firstly, a ratio was calculated between the spectral power of P_{pk1} and P_{log} after one peak had been fitted. It was assumed that the spectrum did not have a dominant frequency component if P_{pk1} contributed less than 50% of power in P_{log}. Secondly, if $f_1 \notin [2, 18]$, or Δ_1 exceeded a threshold TH, the epoch was rejected. In addition, the second peak was discarded by setting A_2 to zero if $\Delta_2 > TH$ or if $f_2 \notin [2, 18]$. The threshold TH was empirically chosen as 1.5 Hz.

Dominant peak correction: The detected frequency components were improved by an intermediate step after obtaining parameters for each peak in the epoch and approximating the curve. Searching for the peak around the estimated frequencies in P_{log} was an effective solution to extract the correct amplitude. Assuming center frequencies f_1 and f_2, peak estimates were shifted towards the positive gradients on P_{log} until a local maximum was achieved. In the event that two peaks were present, and both were updated to the same point, one peak was discarded.

Dominant peak estimation: Having updated the peak parameters of P_{curve}, a power ratio was found between the estimated peak components and other, less dominant, peaks found in the localized segments. The center frequencies f_1 and f_2 were allocated, and a frequency range was found around them based on their center widths Δ_1 and Δ_2. A new spectrum P_{res} was found excluding these frequency ranges:

$$P_{res}(f) = \begin{cases} P(f) & , f \notin R_{peaks} \\ exp(B - C \cdot log(f)) & , f \in R_{peaks} \end{cases}$$

with $R_{peaks} \subseteq [f_{min}, f_{max}]$ being the frequency range around the dominant frequency components. Given the original spectrum P and new spectrum P_{res}, a correlation coefficient was computed:

$$c = 1 - \text{corr}(P, P_{res})$$

If P_{res} was noisy or multiple significant peaks were present, the correlation parameter was low. Conversely, if single peaks contributed most of the power in P_{log}, a high value for c was obtained.

After obtaining the peak components and correlation coefficient for each localized segment, the parameters were grouped into clusters provided that clusters consisting of one peak or smaller in size than 75% of the largest cluster were discarded. Let all the clusters formed be denoted as $\{C_j\}_{j\in\{1,\ldots,N\}}$ with cluster $C_j = \{A_i, f_i, c_i\}_{i\in\{1,\ldots,M_j\}}$ containing the amplitude, frequency and correlation components of the M_j peaks in the cluster. The components were clustered with the function (kmeans scikit-learn v1.1.2). The number of clusters k was found using the elbow method which involves plotting the sum of squared distances between the data points and their cluster centroids (WCSS) against the number of clusters. The optimal number of clusters is the value of k at the "elbow" point, where the decrease in WCSS begins to level off.

Note that the procedure described in the latter was performed for log spectra P_{O1} and P_{O2}. After the peak components were estimated from both log spectra, they were combined. Parameters based on the two largest clusters formed in the set were used to estimate alpha rhythm frequency. In the event that only one cluster was available, we assumed that the recording contained one rhythm. Based on a given cluster, a weighted average of the frequency was calculated:

$$w_i = \frac{c_i}{\sum_k^M c_k}$$

$$Q_f(j) = \sum_i^M f_i w_i$$

with f_i and w_i presenting the estimated frequency and normalized correlation components of the M_j peaks found in a recording. In addition to calculating peak frequencies for a given recording, two further steps were performed. First, if two dominant frequencies were detected, but only one was in the alpha band, the outlying frequency was discarded as it was assumed that it did not belong to the alpha rhythm. Second, if two frequencies were found outside of the alpha range, the one with the largest amplitude was identified as the dominant frequency.

8.2.2 Alpha rhythm reactivity: Using a laplacian montage, let P_{EC} and P_{EO} be the mean occipital power measured in a 0.5 Hz frequency band on the estimated peak frequency for segments with the eyes closed and eyes open state:

$$P_{EC} = \underset{f=\{Q_f \pm .25\}Hz}{mean} [P(c, j, f)], c \in \{O1\}, j \in J_{EC}$$

$$P_{EO} = \underset{f=\{Q_f \pm .25\}Hz}{mean} [P(c, j, f)], c \in \{O1\}, j \in J_{EO}$$

Given the power estimates in the eyes closed and open states, the reactivity in alpha power was quantified as:

$$Q_r = 1 - \frac{P_{EO}}{P_{EC}}$$

8.2.3 Alpha power anterio–posterior gradient: Using a laplacian montage, the mean power in the alpha band was computed as:

$$P_{alpha}(c, j) = \underset{f=\{8\ldots12\}Hz}{mean} [P(c, j, f)]$$

Only segments in the eyes closed state were used and a mean spectrum from these segments was found:

$$P_{EC}(c) = \underset{j\in J_{EC}}{mean} \left[P_{alpha}(c, j)\right]$$

Using channels $\mathbf{c_{ant}} = \{Fp1, Fp2, F7, F8, F3, Fz, F4\}$ and $\mathbf{c_{pos}} = \{T5, T6, P3, P4, Pz, O1, O2\}$, the mean alpha power of the anterior and posterior regions were found following:

$$P_{ant} = \underset{c\in\mathbf{c_{ant}}}{mean} [P_{EC}(c)]$$

$$P_{pos} = \underset{c\in\mathbf{c_{pos}}}{mean} [P_{EC}(c)]$$

With these mean power estimates, the normalised anterio-posterior gradient was defined as:

$$Q_{ap} = \frac{P_{ant}}{P_{ant} + P_{pos}}$$

8.2.4 Interhemispheric asymmetries: Using a laplacian montage, let the channel pairs based be denoted as:

$\mathbf{c_{\{LR\}}}$ = $\{\{Fp1, Fp2\}, \{F7, F8\}, \{F3, F4\}, \{T3, T4\}, \{C3, C4\}, \{T5, T6\}, \{P3, P4\}, \{O1, O2\}\}$

A power ratio for each left-right channel pair $C_{\{LR\}} \in \mathbf{c_{\{LR\}}}$ was found for:

$$LR(C_{\{LR\}}, j, f) = \frac{P\left(C_{\{R\}}, j, f\right) - P\left(C_{\{L\}}, j, f\right)}{P\left(C_{\{R\}}, j, f\right) + P\left(C_{\{L\}}, j, f\right)}$$

By calculating the absolute mean of all segments $j = \{1...M\}$:

$$LR_{avg}(C_{\{LR\}}, f) = \left| \underset{j=\{1...M\}}{mean} \left[LR(C_{\{LR\}}, j, f)\right] \right|$$

and averaging over $f = \{2...12\}$ Hz, a single asymmetry value was obtained for each channel pair:

$$Q_s(C_{\{LR\}}) = \underset{f=\{2...12\}Hz}{mean} \left[LR_{avg}(C_{\{LR\}}, f)\right]$$

8.2.5 Diffuse slow-wave activity: Using a laplacian montage, a mean spectrum was first calculated and the normalised power ratio between $P_{low} = \{2...8\}$ Hz and $P_{wide} = \{2...25\}$ Hz was obtained for channels:

$\mathbf{c_{slow}}$ = $\{F7, F8, F3, F4, Fz, T3, T4, T5, T6, C3, C4, Cz, P3, P4, Pz, O1, O2\}$

Using only segments with the eyes closed state, the mean spectrum over all channels was found:

$$P_{EC}(f) = \underset{C \in \mathbf{c_{slow}}}{mean} \left[\underset{j \in J_{EC}}{mean} \left[P(c, j, f)\right]\right]$$

The discrete power coefficients in the low and wide spectral bands were summed following:

$$P_{low} = \sum_{f=2Hz}^{8Hz} P_{EC}(f)$$

$$P_{wide} = \sum_{f=2Hz}^{25Hz} P_{EC}(f)$$

From these estimates, a ratio was found that quantifies the degree of slow-wave activity:

$$Q_d = \frac{P_{low}}{P_{wide}}$$

8.3 Appendix C: Definition of time-complexity features

For the following features detailed, let $x(t)$ be a time series with $t > 0$ time points.

8.3.1 Higuchi fractal dimension: Higuchi's method for estimating fractal dimension is iterative in nature. The reconstructed sequence of $x(t)$ was computed following:

$$x_m^k = \left\{x(m), x(m+k), ..., x\left(m + \left[\frac{N-m}{k}\right] \cdot k\right)\right\}$$

with k the interval between two adjacent time series and $m = 1, 2, ..., k$ the initial value of the reconstructed sequence of $x(t)$. The box length was calculated for each produced sequence x_m^k:

$$L_m(k) = \frac{\sum_{i=1}^{(N-m)/k} |x(m+ik) - x(m+(i-1)k)| \cdot (N-1)}{[(N-m)/k] \cdot k}$$

The average box length of the sequence was computed as:

$$L(k) = \sum_{m=1}^{k} L_m(k)$$

From this, an useful relation is that the log of the average box length behaves proportional to fractal dimension D times the log of reciprocal k:

$$L(k) \propto \left(\frac{1}{k}\right)^D$$

The linear relation between both parts can be described as a function with slope D presenting Higuchi's fractal dimension. We estimated the slope parameter using a least-squares fit (lstsq numpy v1.22.1) by plotting $\log(L(k))$ as a function of $\log\left(\frac{1}{k}\right)$.

8.3.2 Petrosian fractal dimension: Petrosian's method has similarities with Higuchi's method but differs in a rule to binarize $x(t)$ before estimating the fractal dimension. Suppose that each value was transformed to a binary value following:

$$z_i = \left\{\begin{matrix} 1 & x_i > \bar{x} \\ -1 & x_i \leq \bar{x} \end{matrix} \quad i = 1, 2, ..., n \right\}$$

with $\bar{x}$ the mean of $x(t)$ and x_i a value in $x(t)$. Each binary value was stored in sequence z_i and the total number of adjacent value changes was computed:

$$N_\Delta = \sum_{i=1}^{N-2} \frac{z_{i+1} - z_i}{2}$$

The fractal dimension D was found using the relation:

$$D = \frac{log_{10}^N}{log_{10}^N + log_{10}\left(\frac{N}{N+0.4N_\Delta}\right)}$$

8.3.3 Hurst exponent: Time series $x(t)$ of length n was divided into a number of shorter series of lengths $n = n, n/2, n/4, ...$, and the average rescaled range was calculated for each value of n. To do this, first the mean of $x(t)$ was computed,

denoted as $\bar{x}$. Then a separate mean-adjusted series was created and subsequently used to calculate the cumulative deviate series $Z(t)$:

$$Z(t) = \sum_{i=1}^{t} (x_i - \bar{x})$$

The range $R(n)$ was found by subtracting the maximum and minimum value from $Z(t)$ and the standard deviation $S(n)$ was found following:

$$R(n) = \max(z_1, z_2, ..., z_n) - \min(z_1, z_2, ..., z_n)$$

$$S(n) = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})^2}$$

By taking the average of the rescaled range $R(n)/S(n)$ over all partial series of length n, a straight line was fitted describing the power law behavior:

$$\frac{R(n)}{S(n)} \propto C(n)^H$$

with C a constant and n the time span of the observation. Hurst's exponent was estimated by plotting $\log[R(n)/S(n)]$ as a function of $\log n$. The slope parameter was estimated using a least-squares fit (lstsq numpy v1.22.1).

8.3.4 ***Largest Lyapunov exponent:*** The first step involves constructing an attractor dynamics from $x(t)$. The reconstructed trajectory X can be expressed as a matrix with each row a phase-space vector. The delayed sequence of $x(t)$ was constructed using the following criteria:

$$M = \{X(i), X(i + J), ..., X(i + (m - 1) \cdot J)\}$$

with J the lag or reconstruction delay, and m the embedding dimension. J was computed using the FFT and the final parameters used were $J = 4$ and $m = 10$.

Following the reconstruction of the series dynamics, the nearest neighbors of each point on the trajectory were identified. The closest neighbor $X_{\hat{j}}$ was selected by searching for a point that minimizes distance from a specific reference point X_j:

$$d_j(0) = \min_{X_{\hat{j}}} \|X_j - X_{\hat{j}}\|$$

with $d_j(0)$ the initial distance from the jth point to its nearest neighbor. $\|...\|$ denotes the Euclidean norm. As an additional step we required the nearest neighbors to be separated by a longer duration than the time series mean duration:

$$|j - \hat{j}| > \text{mean period}$$

Each pair of neighbors was considered as nearby for initial conditions for a variety of trajectories. Using the mean rate of separation between the nearest neighbors, the largest Lyapunov exponent was estimated. As far as the method for calculating λ_1 is concerned, according to Rosensteins' method, the largest Lyapunov exponent is defined as:

$$d(t) = Ce^{\lambda_1 t}$$

with $d(t)$ the average divergence at time t and C a constant that normalizes the initial separation (Rosenstein et al., 1993). We assumed that the jth pair of neighbors diverge approximately at the rate by the largest Lyapunov exponent:

$$d_j(t) \propto C_j e^{\lambda_1 (i\Delta t)}$$

The equation in the latter presents a set of approximately parallel lines $j = 1, 2, ..., N$, each with a slope roughly proportional to λ_1. The slope parameter (i.e., the largest Lyapunov exponent) was estimated from the average of all parallel lines using a least-squares fit (lstsq numpy v1.22.1).

8.3.5 ***Phase locking value:*** Consider a pair of time series $x_i(t)$ for $i = 1, 2$. Let the output of the Hilbert transform be denoted as the analytical signal $z_i(t)$:

$$z_i(t) = A_i(t) \cdot e^{-j\phi_i(t)}$$

The instantaneous phase $\phi_i(t)$ is the angle between the real and the imaginary parts of the Hilbert analytical signal. After the analytical signals were obtained (hilbert scipy v1.9.1), the relative phase was computed:

$$\Delta\phi(t) = arg\left(\frac{z_1(t)z_2(t)}{||z_1(t)|| \cdot ||z_2(t)||}\right)$$

with the argument of a complex number z, denoted $arg(z)$, the angle between the positive real axis and the line joining the origin. $\|...\|$ denotes the Euclidean norm. A phase close to 1 describes synchronized signals, with 0 reflecting no phase synchrony.

8.4 Appendix D: Decoding results

Table 4 Bootstrapped decoding results for the multi- and single-channel decoding experiments. Performance outcomes are displayed for various combinations of features and classifiers. The results are generated by computing features at varying lengths of the signal data. The best performing models are emphasized for each signal data duration used for feature extraction. The combined features consist of the most effective spatial features, i.e., PWR, COGx, and COGy, optimized for maximum performance.

Reduced signal data	2min		4min		8min		16min	
	Spec (%)	Sens (%)	Spec (%)	Sens (%)	Spec (%)	Sens (%)	Spec (%)	Sens (%)
Model	(mean, std)	(mean, std)	(mean, std)	(mean, std)	(mean, std)	(mean, std)	(mean, std)	(mean, std)
Cropped-RF	**(79.1, 1.04)**	**(75.2, 1.14)**	**(81.0, 0.94)**	**(76.0, 1.03)**	(83.9, 1.03)	(76.5, 1.08)	(80.8, 1.21)	(76.8, 1.23)
PWR-RF	(77.9, 1.01)	(75.9, 1.13)	(79.9, 1.04)	(75.4, 1.11)	**(85.0, 1.08)**	**(76.4, 1.09)**	**(80.9, 1.18)**	**(76.9, 1.26)**
COGx-RF	(73.9, 1.14)	(73.9, 1.16)	(75.9, 1.14)	(75.8, 1.17)	(80.9, 0.86)	(76.9, 0.98)	(79.0, 0.92)	(76.0, 1.14)
COGy-RF	(75.3, 1.24)	(74.3, 1.21)	(76.9, 0.95)	(76.9, 1.17)	(82.1, 1.05)	(76.1, 1.06)	(78.9, 1.13)	(75.9, 1.15)
mNNC-RF	(67.9, 1.0)	(68.9, 1.10)	(69.8, 0.96)	(70.8, 1.08)	(74.0, 1.21)	(71.0, 1.23)	(73.0, 1.15)	(70.0, 1.27)
Combined-RF	(77.9, 1.12)	(70.1, 1.29)	(78.9, 1.02)	(71.2, 1.29)	(81.0, 1.20)	(72.0, 1.21)	(79.0, 1.12)	(71.5, 1.18)
Cropped-SVC	(76.9, 1.05)	(73.0, 1.14)	(77.8, 1.13)	(74.8, 1.27)	(81.2, 1.26)	(76.2, 1.19)	(79.9, 1.32)	(74.9, 1.28)
PWR-SVC	(76.2, 1.05)	(75.2, 0.98)	(76.1, 0.91)	(74.1, 1.00)	(81.0, 1.05)	(76.2, 1.09)	(79.0, 1.07)	(76.0, 1.10)
COGx-SVC	(72.9, 0.91)	(73.9, 1.00)	(74.0, 1.09)	(74.0, 1.20)	(78.9, 0.99)	(76.0, 1.00)	(76.9, 1.25)	(75.0, 1.30)
COGy-SVC	(74.2, 1.02)	(73.0, 1.02)	(74.8, 1.24)	(75.8, 1.17)	(77.8, 1.25)	(76.8, 1.31)	(76.9, 1.38)	(74.9, 1.33)
mNNC-SVC	(66.0, 1.19)	(68.0, 1.20)	(66.9, 1.03)	(69.8, 1.12)	(78.9, 1.07)	(69.9, 1.17)	(69.9, 1.27)	(67.9, 1.31)
Combined-SVC	(75.2, 1.25)	(69.2, 1.25)	(76.1, 1.07)	(70.1, 1.32)	(77.0, 1.14)	(70.9, 1.29)	(76.1, 1.08)	(70.1, 1.31)
Cropped-LR	(78.0, 1.01)	(76.1, 1.20)	(79.1, 1.11)	(75.1, 1.10)	(81.9, 1.13)	(75.8, 1.22)	(80.9, 1.19)	(75.9, 1.20)
PWR-LR	(76.9, 1.15)	(74.9, 1.21)	(77.8, 1.09)	(75.8, 1.19)	(83.2, 1.12)	(77.3, 1.21)	(79.8, 1.18)	(75.8, 1.26)
COGx-LR	(73.9, 1.06)	(75.0, 1.06)	(74.9, 1.01)	(75.0, 1.13)	(79.1, 1.24)	(76.1, 1.27)	(80.1, 1.14)	(75.2, 1.20)
COGy-LR	(75.0, 1.06)	(74.0, 1.10)	(76.1, 1.03)	(77.1, 1.04)	(79.9, 0.98)	(76.0, 0.98)	(79.1, 0.99)	(75.1, 1.21)
mNNC-LR	(67.9, 1.13)	(67.0, 1.25)	(68.1, 1.15)	(71.1, 1.24)	(71.0, 1.07)	(70.0, 1.18)	(71.0, 1.12)	(67.5, 1.24)
Combined-LR	(76.0, 1.17)	(70.8, 1.18)	(77.0, 1.23)	(71.0, 1.21)	(79.0, 0.96)	(71.8, 1.18)	(76.9, 1.15)	(70.6, 1.23)